

Control work 1 · Algebraic fractions

The first assessment of the quarter, followed by a full lesson on the mistakes it produced.

LESSON 15–16

2 × 40 MIN

ALGEBRA 8, §§1–4

STAGE 9 · UNIT 2 CHECK

LESSON CLOCK

2'

36'

2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess: permissible values, cancelling, common denominators, addition and subtraction.
- Work through the errors the paper reveals, by type rather than by question.
- Re-solve every task that was lost, correctly and in full.

TEXTBOOK

National	Revision of §§1–4
Cambridge	Learner’s Book pp. 21–54
Workbook	Workbook Unit 2

01

Explanation

Lesson 15 — the paper (40 minutes)

Two variants of seven tasks, printed below as the Easy and Medium practice sets. Both variants carry the same marks and cover the same four skills:

- tasks 1–2 · permissible values (§1)
- tasks 3–4 · cancelling a fraction (§2)
- task 5 · the lowest common denominator (§3)
- tasks 6–7 · addition and subtraction (§4)

Marking: 2 marks per task, 14 in total. A mark is lost for a missing permissible-value condition even when the algebra is right — say this before the paper starts, not after.

Lesson 16 — work on mistakes (40 minutes)

Do not go through the paper question by question. Sort the errors into the four types below, deal with each type once, then have every learner re-solve only their own lost tasks.

THE FOUR ERRORS THAT ACCOUNT FOR ALMOST EVERYTHING

1. **Cancelling a term instead of a factor** — $\frac{x+5}{5}$ written as x .
2. **The missing bracket in a subtraction** — $2x - x + 2$ instead of $2x - (x + 2)$.
3. **The permissible values dropped**, or taken from the *simplified* denominator instead of the original one.
4. **Multiplying the denominators together** without factorising, then failing to cancel back.

The Hard set below is built entirely from these four errors: each task is a piece of wrong working to be found and repaired. That is a better use of the hour than a second full paper.

02 Worked examples

EXAMPLE 1 Find and repair the error: $\frac{x+4}{4} = x$

- ① The 4 on top is a **term** — it is added to x , not multiplied by it.
Only factors cancel.

- ② Write the numerator as a product to test it: $x + 4$ has no factor 4.
So there is nothing to cancel.

- ③ $\frac{x+4}{4} = \frac{x}{4} + 1$
You may split it, but you may not cancel it.

ANSWER

The expression does not simplify to x . A correct alternative form is $\frac{x}{4} + 1$.

EXAMPLE 2 Find and repair the error: $\frac{3x}{x^2 - 1} - \frac{1}{x - 1} = \frac{3x - 1}{x^2 - 1}$

① The second fraction was not built up to the LCD.
Its denominator is $(x - 1)$, not $(x - 1)(x + 1)$.

② *additional factor $(x + 1)$*
So the numerator becomes $(x + 1)$, not 1.

③ $\frac{3x - (x + 1)}{(x - 1)(x + 1)} = \frac{2x - 1}{x^2 - 1}$
Both errors — the factor and the bracket — corrected.

ANSWER

$$\frac{2x - 1}{x^2 - 1}, \quad x \neq \pm 1$$

03 Interactive model

Use this in lesson 16: put a piece of wrong working on the board and let the class find the error before you reveal it.

Find the mistake

Claimed: $\frac{x + 5}{5} = x$

- ① The 5 in the numerator is added, not multiplied.

Cancelling
needs
a
common
factor.

- ② Test it with a number: at

$x = 5$, the left side is

, the right side is 5.

$$\frac{10}{5} = 2$$

One
counter-
example
settles
it.

- ③ Correct: the fraction does not simplify. It may be split as

$$\frac{x}{5} + 1$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Variant	Variant	Вариант
Mark	Ball	Балл
Check the answer	Javobni tekshirish	Проверить ответ
Error	Xato	Ошибка
Correct	To'g'ri	Верно
Counter-example	Qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

A learner writes $\frac{x+7}{7} = x$. The error is:

 an arithmetic slip

 cancelling a term instead of a factor

 the permissible values

 nothing — it is correct

In $\frac{4x}{x^2-1} - \frac{1}{x-1}$ the numerator after combining is:

 $4x - 1$
 $4x - x - 1$
 $4x - (x + 1)$
 $4x + x + 1$

After simplifying $\frac{x^2-4}{x-2}$ to $x+2$, the condition is:

 $x \neq -2$
 $x \neq 2$
 no condition

 $x \neq 0$

The lowest common denominator of $\frac{1}{x^2 - 4}$ and $\frac{1}{x + 2}$ is:

$$(x^2 - 4)(x + 2)$$

$$x^2 - 4$$

$$(x + 2)^2$$

$$x - 2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** State the permissible values of $\frac{x+2}{x-6}$

ANSWER $x \neq 6$

2. **Task 2.** State the permissible values of $\frac{4}{x^2 - 16}$

ANSWER $x \neq \pm 4$

3. **Task 3.** Cancel $\frac{15ab}{20a}$

ANSWER $\frac{3b}{4}, a \neq 0$

4. **Task 4.** Cancel $\frac{x^2 - 16}{x + 4}$

ANSWER $x - 4, x \neq -4$

5. **Task 5.** Find the LCD of $\frac{1}{x^2 - 4}$ and $\frac{1}{x - 2}$

ANSWER $(x - 2)(x + 2)$

6. **Task 6.** $\frac{5}{x} + \frac{2}{x}$

ANSWER $\frac{7}{x}$

7. **Task 7.** $\frac{1}{x - 3} + \frac{1}{x + 3}$

ANSWER $\frac{2x}{x^2 - 9}$

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** State the permissible values of $\frac{x-1}{x+5}$

ANSWER $x \neq -5$

2. **Task 2.** State the permissible values of $\frac{3}{x^2 - 5x}$

ANSWER $x \neq 0, x \neq 5$

3. **Task 3.** Cancel $\frac{18xy}{24y}$

ANSWER $\frac{3x}{4}, y \neq 0$

4. **Task 4.** Cancel $\frac{x^2 - 8x + 16}{x^2 - 16}$

ANSWER $\frac{x-4}{x+4}, x \neq \pm 4$

5. **Task 5.** Find the LCD of $\frac{1}{x^2 + 3x}$ and $\frac{1}{x + 3}$

ANSWER $x(x + 3)$

6. **Task 6.** $\frac{9}{2a} - \frac{3}{2a}$

ANSWER $\frac{3}{a}$

7. **Task 7.** $\frac{3x}{x^2 - 1} - \frac{1}{x - 1}$

ANSWER $\frac{2x - 1}{x^2 - 1}$

Hard · stretch and reason

7 PROBLEMS

1. Find the error: $\frac{x+6}{6} = x$

ANSWER A term was cancelled. The expression equals $\frac{x}{6} + 1$; test with $x = 6$.

2. Find the error: $\frac{2x}{x^2-4} - \frac{1}{x-2} = \frac{2x-1}{x^2-4}$

ANSWER The second fraction was not built up. Correct answer $\frac{1}{x+2}$.

3. Find the error: $\frac{x^2-25}{x-5} = x+5$ for every x

ANSWER True only for $x \neq 5$; the condition comes from the original denominator.

4. Find the error: LCD of $\frac{1}{x^2-9}$ and $\frac{1}{x-3}$ is $(x^2-9)(x-3)$

ANSWER Not lowest. $LCD = x^2 - 9$.

5. Find the error: $\frac{1}{x} + \frac{1}{y} = \frac{1}{x+y}$

ANSWER Denominators are never added. Correct: $\frac{x+y}{xy}$; check with $x = y = 2$.

6. Find the error: $\frac{a-b}{b-a} = 1$

ANSWER The brackets are reversed: $b-a = -(a-b)$, so the value is -1 .

7. Find the error: $\frac{3}{x-2} - \frac{3}{x+2} = 0$

ANSWER Different denominators cannot cancel. Correct: $\frac{12}{x^2-4}$.

07 Homework

Homework after the work-on-mistakes lesson

Re-solve, in full, every task you lost marks on. Then these three.

1. Cancel $\frac{x^2 - 7x + 12}{x^2 - 9}$ and state the permissible values

2. $\frac{2}{x - 4} - \frac{2}{x + 4}$

3. Write out, in your own words, the four errors listed in this lesson and one example of each.